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Maximal Strength in Relation to Force and Velocity Patterns During Countermovement Jumps

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Maximal strength is important for the performance of dynamic athletic activities, such as countermovement jumps (CMJ). Although measures of maximal strength appear related to discrete CMJ variables, such as peak ground reaction forces (GRF) and center-of-mass (COM) velocity, knowledge about the association between strength and the time series patterns during CMJ will help characterize changes that can be expected in dynamic movement with changes in maximal strength. ***Purpose:*** To investigate the associations between maximal strength and GRF and COM velocity patterns during CMJ. ***Methods:*** Nineteen female college lacrosse players performed 3 maximal-effort CMJs and isometric midthigh pull. GRF and COM velocity time series data from the CMJ were time normalized and used as inputs to principal-components analyses. Associations between isometric midthigh pull peak force and CMJ principal-component scores were investigated with a correlational analysis. ***Results:*** Isometric midthigh pull peak force was associated with several GRF and COM velocity patterns. Correlations indicated that stronger players exhibited a GRF pattern characterized by greater eccentric-phase rate of force development, greater peak GRF, and a unimodal GRF profile ($P = .016$). Furthermore, stronger athletes exhibited a COM velocity pattern characterized by higher velocities during the concentric phase ($P = .004$). ***Conclusions:*** Maximal strength is correlated to specific GRF and COM velocity patterns during CMJ in female college lacrosse athletes. Since maximal strength was not correlated with discrete CMJ variables, the patterns extracted via principal-components analyses may provide information that is more beneficial for performance coaches and researchers.

Keywords: biomechanics, ground-reaction forces, principal-component analysis, isometric midthigh pull

The importance of maximal strength for athletic performance is well established, and isometric maximal strength appears related to dynamic performance of various athletic tasks, such as the countermovement jump (CMJ).[1,2] One commonly used assessment of isometric maximal strength is the isometric midthigh pull (IMTP), which is used by practitioners because it is safe, easy to administer, highly reliable within and between sessions, and minimally fatiguing to athletes.[3] In addition, body positioning during the IMTP closely replicates the positioning observed during dynamic movements that require triple extension of the lower-extremity joints, like the CMJ.[4] It is, therefore, not a surprise that numerous studies have found correlations between IMTP and dynamic performance measures.[5–10] Specifically, stronger individuals exhibit superior CMJ performance when compared with weaker athletes. Nuzzo et al[7] found that IMTP peak force was correlated to CMJ peak power and that IMTP relative peak force was correlated to CMJ height in collegiate football players and track athletes. Similarly, separate studies from West et al,[10] Khamoui et al,[6] and Thomas et al[9] on professional rugby league players, recreationally trained men, and young female athletes, respectively, all found associations between IMTP relative peak force and jump height. However, relationships between isometric strength and discrete measures of jump performance are not always consistent. For example, Thomas et al[8] found that IMTP peak force correlated with CMJ peak force but not with CMJ height. In addition, there were no correlations between IMTP relative peak force and CMJ variables.

Because of equivocal findings in the literature, researchers have questioned the usefulness of discrete variables in the context of CMJ testing.[11,12] Importantly, how these variables are identified may account for some inconsistencies in reported relationships between IMTP peak force and discrete CMJ measures. Indeed, extracting discrete variables from time series data may limit insights into performance relationships because the temporal sequencing or time-varying patterns during dynamic movements are often neglected. Consequently, important biomechanical information relevant to movement performance may be ignored. For example, with respect to the analysis of force–time curves, combining data from trials that include both unimodal and bimodal ground reaction force (GRF) curves could be problematic.[13] Some authors have tried to address the shortcoming of discrete variable analysis with temporal phase analysis, exploring differences in kinetic and kinematic variables throughout CMJ.[14,15] However, although Cormie et al[14] used temporal phase analysis to purport that training status influences force–time and velocity–time CMJ curves, the temporal phase analysis method simply makes pairwise comparisons of every single discrete point of the time series data, which may cause inadvertent statistical problems related to type I errors.

Considering the impulse–momentum theorem and degeneracy of the neuromuscular system, athletes can use different movement strategies or patterns to achieve similarities in outcomes (eg, jump height). If the overarching goal of CMJ testing is to assess the state of the neuromuscular system, analyzing the time series data or dynamics across an entire movement trial may better identify or characterize movement strategies or patterns. In turn, practitioners should gain valuable information about training status and performance outcomes that is more definitive than discrete variable

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analysis. One alternative to selecting discrete data at distinct time points is to employ analyses that can use entire waveforms as inputs, such as principal components analysis (PCA), because they can facilitate a better understanding of the dynamics of time series data.[16] The effectiveness of PCA with CMJ data has been previously demonstrated in that it can be effective in parsing out between-sport differences in the time series GRF.[17] Thus, conducting a PCA on dynamic tasks like the CMJ allows for extraction of patterns within the force–time and velocity–time curves, providing valuable insight into GRF and center of mass (COM) velocity patterns. Exploring the correlations between IMTP peak force and GRF and COM velocity patterns during CMJ would be valuable as such information would provide practitioners with much better insight about the relationships between strength capacities and movement characteristics during jumping tasks. Thus, the purpose of this study is to investigate the associations between maximal isometric strength, via IMTP, and GRF and COM velocity patterns during CMJ. Exploring and characterizing these associations is necessary before forming directional hypotheses and will lay the foundation for more specific experimental paradigms that focus on investigating changes in CMJ movement patterns due to training, fatigue, or athlete feedback.

Methods

Subjects

Nineteen female Division I lacrosse athletes (age = 19.5 [1.2] y; body mass = 70.0 [9.4] kg; height = 166.8 [4.1] cm) volunteered for this study. All players participated in a yearly training program that included dynamic and isometric exercises. The study was approved by Marquette University's Institutional Review Board, and all players provided written informed consent.

Methodology

Data collection occurred at the beginning of an off-season training period, and testing for each athlete was completed within one session. Upon arrival for testing, players performed a standardized dynamic warm-up that included basic callisthenic exercises. Participants then performed 2 submaximal CMJ before performing up to 3 maximal effort CMJ, followed by 2 submaximal effort IMTP and 3 maximal effort IMTP with participants positioned based on previous recommendations.[3] Before all maximal effort test trials, athletes were instructed to perform each exercise with maximal effort as they would during regular training sessions with their performance coaches. Verbal encouragement was provided during each test trial with approximately 1 minute of rest between trials.

During the CMJ, GRF data were collected with 2 force plates (Advanced Mechanical Technology, Inc, Watertown, MA) at 1000 Hz. Alternatively, during the IMTP, GRF data were collected with only one force plate (Kistler Instrument Corp, Novi, MI) at 1000 Hz. All GRF data were smoothed with a fourth-order zero-lag Butterworth filter of 15 Hz, selected based on residual analyses and effects on the signal-to-noise ratio of the first-order derivative of the GRF. Filtered GRF data from the 2 force plates during the CMJ trials were summed into a single GRF vector. All GRF data were normalized to the athletes' body mass. Data from the CMJ were used to calculate COM velocity throughout the movement with a custom-written script in MATLAB (The MathWorks Inc, Natick, MA). In addition, the GRF and COM velocity time series data were time normalized to 101 data points to represent 100% of the CMJ movement phase wherein movement onset was defined as the point where the filtered GRF fell below 95% of body weight and the end of the movement was defined as the point where the filtered GRF fell below 10 N.

The time-normalized GRF and COM velocity time series data from the CMJ were each used as input to a separate PCA, which was also performed in MATLAB. Input data were centered, and the eigenvector decomposition algorithm was used to extract principal components (PC) and the variance accounted for by each PC. The PC scores were then calculated for each CMJ trial. Within-subject averages of PC scores and discrete outcomes were calculated from data of all successful CMJ and IMTP trials and used for all statistical analyses.

Statistical Analysis

Associations between peak IMTP force and discrete CMJ variables, as well as CMJ PC scores, were investigated with a correlational analysis. To investigate relationships between force and velocity patterns, a correlation analysis was also performed between force–time and velocity–time PC scores. Due to the violation of normality, Spearman ρ correlations were used. Bootstrapping with 1000 replicate samples was used to increase the robustness of the analyses and allow for reporting of bias corrected and accelerated 95% confidence intervals. All statistical analyses were performed with SPSS (version 26.0; IBM Corp, Armonk, NY).

Results

From the 19 participants, 51 CMJ and 57 IMTP trials were analyzed (Table 1). IMTP peak force was not correlated to any discrete CMJ variables (Table 2).

The PCA of the force–time data indicated that 3 PC explained 93.0% of the variance in GRF curves during the CMJ (Figure 1). The 3 extracted patterns captured notable variations in the GRF curves (Figure 2). The first pattern (PC1) identified variations in the GRF amplitude during the unweighing phase and in the peak GRF, where greater unweighing was also associated with greater peak GRF during the eccentric phase (ie, pattern of [+] PC scores). The second pattern (PC2) captured differences in eccentric-phase rate of force development (RFD), the overall peak GRF, and the presence of a bimodal GRF profile, where lesser overall GRF were also associated with a bimodal GRF profile and greater overall GRF were associated with a unimodal profile (ie, [−] PC scores and [+] PC scores, respectively). Finally, the third pattern captured variation in the relative duration of the unweighing phase and in the GRF during the latter part of the concentric phase, where a shorter relative unweighing phase was also associated with greater concentric plateau (ie, pattern of [+] PC scores).

The velocity–time PCA indicated that 3 PC could explain 92.4% of the variance in COM velocity curves during the CMJ (Figure 3). The first pattern (PC1; Figure 4) identified differences in the velocity during the unweighing and eccentric phases of the CMJ, suggesting that higher velocities during unweighing is followed by higher velocities during the eccentric phase (ie, pattern of [+] PC scores). Next, the second pattern (PC2; Figure 4) appeared to capture variation in the peak velocity during the concentric phase of the jump. Finally, the third component extracted from the velocity PCA only accounted for 6.6% of the variance in the data and seemed to highlight differences in peak velocity just prior to take off.

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Table 1 Descriptive Statistics of Discrete CMJ and IMTP Variables

Outcome Variable	Mean (SD)	Minimum	Maximum
CMJ peak velocity, m/s	2.31 (0.17)	1.81	2.52
CMJ peak force, N	1607 (215)	1194	2142
CMJ peak rate of force development, N/s	10,324 (4389)	5641	24,794
CMJ peak power, W	2783 (290)	2344	3655
CMJ height, cm	27.36 (3.72)	16.61	32.45
IMTP normalized peak force, N/kg	31.67 (3.41)	26.19	39.89

Abbreviations: CMJ, countermovement jump; IMTP, isometric midthigh pull.

Table 2 Spearman (ρ) Correlations and 95% CIs Between IMTP Peak Force and Discrete CMJ Variables

CMJ variable	ρ	*P*	95% CI	
Jump height	.308	.199	–.173	.733
Peak force	.233	.336	–.308	.743
Peak RFD	.339	.156	–.146	.763
Peak power	.172	.482	–.365	.668

Abbreviations: CI, confidence interval; CMJ, countermovement jump; IMTP, isometric midthigh pull; RFD, rate of force development.

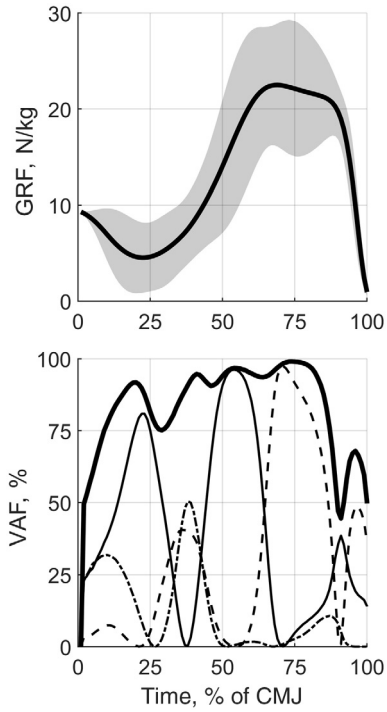

Figure 1 — Ensemble average (mean ± SD) for body-mass-normalized GRF time series data during CMJ (top panel). Variance accounted for VAF of the GRF time series data by the principal components (bottom panel). Total VAF (black line), VAF by PC1 (thin line), VAF by PC2 (dashed line), and VAF by PC3 (dotted line). CMJ indicates countermovement jump; GRF, ground-reaction forces; PC, principal component; VAF, variance accounted for.

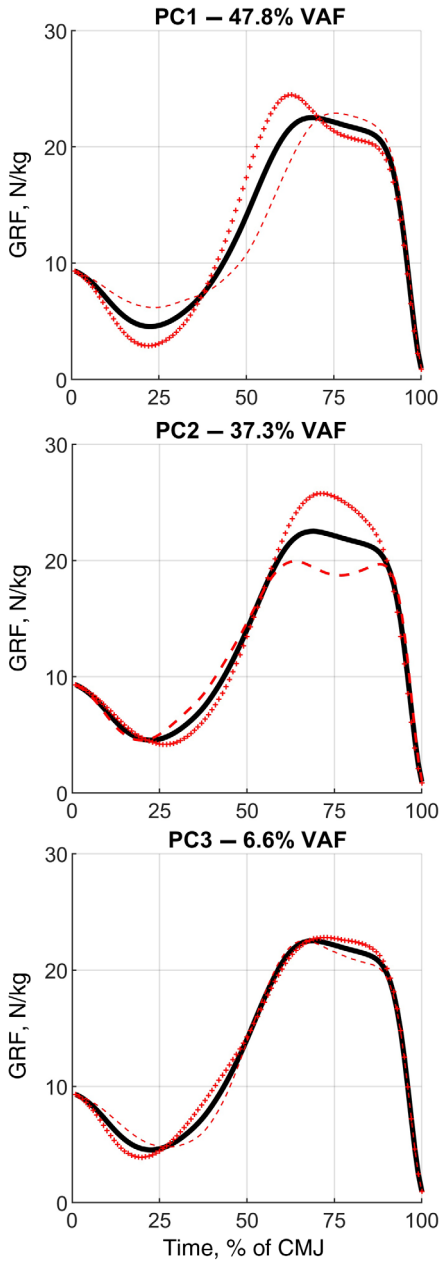

Figure 2 — Ensemble average (solid line) of body-mass-normalized GRF time series data during the CMJ along with the effects of each PC on the shape of the GRF curves (effects of positive and negative PC scores are illustrated with + and – symbols, respectively). CMJ indicates countermovement jump; GRF, ground-reaction forces; PC, principal component; PC, principal componen; VAF, variance accounted for.

Discrete IMTP and CMJ variables were significantly correlated with 2 GRF patterns (PC2 and PC3; Table 3). Whereas PC2 was significantly correlated with IMTP peak force and CMJ force, RFD, and power, PC3 was correlated to CMJ height. Interestingly,

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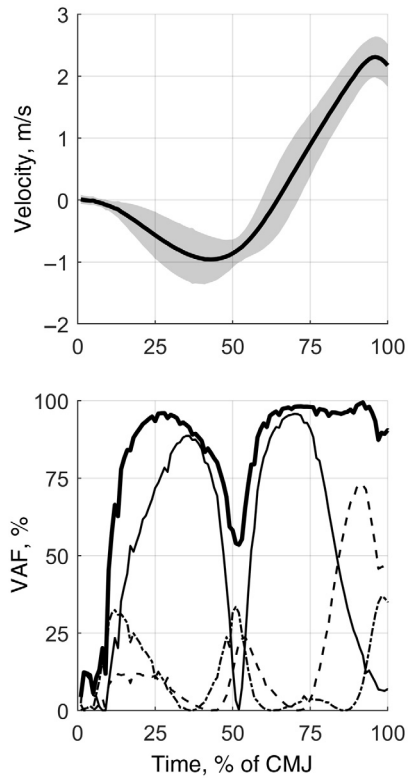

Figure 3 — Ensemble average (mean ± *SD*) for center-of-mass velocity time series data during CMJ (top panel). VAF of the center-of-mass velocity time series data by the PCs (bottom panel). Total VAF (black line), VAF by PC1 (thin line), VAF by PC2 (dashed line), and VAF by PC3 (dotted line). CMJ indicates countermovement jump; PC, principal component; VAF, variance accounted for.

the pattern that accounted for the most variation in the data (PC1, Figure 2) was not correlated to any discrete parameters. As with the GRF patterns, correlation analysis between velocity–time PC and discrete performance variables resulted in multiple significant associations (Table 4). Interestingly, all the COM velocity PC scores were correlated with CMJ height. Furthermore, PC2 was positively correlated with CMJ power, RFD, and maximal strength (IMTP peak force).

Significant associations were also found between the PC scores derived separately from the force–time and velocity–time curves ($P < .001$; Table 5).

Discussion

To our knowledge, this is the first study to examine correlations between maximal isometric strength and GRF time series patterns during the CMJ. The correlation analysis showed that stronger players exhibited a GRF pattern (PC2) characterized by greater eccentric phase RFD, greater peak GRF, and a unimodal GRF profile. The PC scores for this pattern were also correlated with discrete peak force, RFD, and power during the CMJ, which supported the interpretation of the captured GRF pattern. The observed correlation between maximal isometric strength and GRF patterns is similar to results from previous research that illustrated relationships between maximal strength and dynamic performance.[1,2] However, IMTP peak force did not correlate with any of the discrete CMJ parameters, most notably CMJ height,

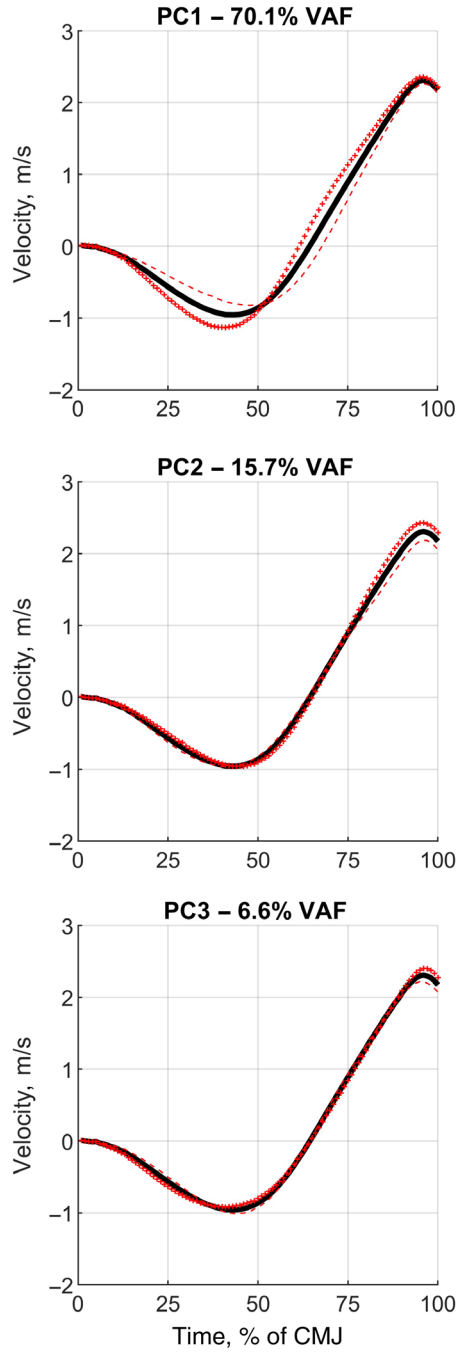

Figure 4 — Ensemble average (solid line) of center-of-mass velocity time series data during the CMJ along with the effects of each PC on the shape of the center-of-mass velocity curves (effects of positive and negative PC scores are illustrated with +/− symbols, respectively). CMJ indicates countermovement jump; PC, principal component; VAF, variance accounted for.

which conflicts with other studies.[5–7,9,10] In the current study, only the third GRF pattern correlated with CMJ height. This pattern captured variations in the relative duration of the unweighing phase as a percentage of the CMJ time and a plateau of the GRF during the concentric phase. The correlation results suggested that athletes who spent less relative time in the unweighing phase and exhibited greater GRF across the

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Table 3 Spearman (ρ) Correlations and 95% CIs Between GRF PC Scores and Discrete IMTP and CMJ Variables

PC score	Variable	ρ	*P*	95% CI	
PC1	IMTP peak force	.063	.797	–.459	.551
	CMJ height	.306	.202	–.198	.694
	CMJ peak force	–.244	.314	–.619	.202
	CMJ peak RFD	.316	.188	–.193	.708
	CMJ peak power	–.260	.283	–.715	.286
PC2	IMTP peak force	**.544**	**.016**	**.111**	**.830**
	CMJ height	.159	.516	–.338	.681
	CMJ peak force	**.540**	**.017**	**.008**	**.885**
	CMJ peak RFD	**.712**	**.001**	**.367**	**.876**
	CMJ peak power	**.460**	**.048**	**.033**	**.733**
PC3	IMTP peak force	.167	.495	–.380	.702
	CMJ height	**.566**	**.012**	**.193**	**.783**
	CMJ peak force	–.472	.041	–.800	.065
	CMJ peak RFD	–.037	.881	–.554	.476
	CMJ peak power	–.198	.552	–.602	.364

Abbreviations: CI, confidence interval; CMJ, countermovement jump; GRF, ground-reaction force; IMTP, isometric midthigh pulls; PC, principal component. Note: Significant ρ values with $P \leq .05$ and 95% CI that do not contain 0 are presented in bold.

Table 4 Spearman (ρ) Correlations and 95% CIs Between COM Velocity PC Scores and Discrete IMTP and CMJ Variables

PC score	Variable	ρ	*P*	95% CI	
PC1	IMTP peak force	–.054	.825	–.509	.414
	CMJ height	**.529**	**.020**	**.110**	**.804**
	CMJ peak force	–.456	.050	–.794	.051
	CMJ peak RFD	.044	.858	–.438	.531
	CMJ peak power	–.317	.186	–.681	.201
PC2	IMTP peak force	**.634**	**.004**	**.242**	**.883**
	CMJ height	**.622**	**.004**	**.197**	**.900**
	CMJ peak force	.392	.097	–.196	.849
	CMJ peak RFD	**.532**	**.019**	**.115**	**.786**
	CMJ peak power	**.545**	**.016**	**.234**	**.741**
PC3	IMTP peak force	.008	.974	–.556	.595
	CMJ height	**.657**	**.002**	**.352**	**.829**
	CMJ peak force	–.498	.030	–.851	.101
	CMJ peak RFD	–.365	.124	–.781	.197
	CMJ peak power	.037	.881	–.443	.545

Abbreviations: CI, confidence interval; CMJ, countermovement jump; COM, center of mass; IMTP, isometric midthigh pulls; PC, principal component. Note: Significant ρ values with $P \leq .05$ and 95% CI that do not contain 0 are presented in bold.

concentric phase were able to reach greater CMJ heights. This pattern, however, was not related to maximal strength, which aligned with the findings of Thomas et al[8] in that peak IMTP force did not correlate with CMJ height. One of limitations of using discrete CMJ variables is the loss of important biomechanical data relevant to performance.[11,12] For example, when examining dynamic (CMJ) performance in relation to isometric (IMTP) performance through the analysis of only discrete peak values, the temporal aspects of force production and their associations with performance are largely ignored. PCA, however, allows for analysis of patterns in both magnitude and timing of GRF during the CMJ. The lack of significant correlations between peak IMTP force and discrete CMJ variables in the present study underscore these issues, suggesting that the choice of analysis technique (PCA vs discrete variable analysis) can significantly influence the results and their interpretation.

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Table 5 Spearman (ρ) Correlations and 95% CIs Between GRF PC Scores and COM Velocity PC Scores

PC score					
GRF	**COM velocity**	ρ	***P***	**95% CI**	
PC1	PC1	**.883**	**<.001**	**.623**	**.984**
	PC2	.114	.642	−.382	.546
	PC3	−.153	.533	−.569	.273
PC2	PC1	−.243	.316	−.691	.204
	PC2	**.813**	**<.001**	**.539**	**.945**
	PC3	−.189	.439	−.753	.470
PC3	PC1	.390	.099	−.058	.701
	PC2	.176	.470	−.372	.715
	PC3	**.783**	**<.001**	**.519**	**.915**

Abbreviations: CI, confidence interval; COM, center of mass; GRF, ground reaction force; PC, principal component. Note: Significant ρ values with $P \leq .05$ and 95% CI that do not contain 0 are presented in bold.

The correlation analysis also showed that stronger players exhibited a COM velocity pattern (PC2) characterized by greater peak velocity during the concentric phase of the jump. Notably, the PC scores for this pattern were also correlated with discrete peak RFD and power and CMJ jump height. It, therefore, appears that athletes with greater maximal isometric strength reached higher velocities during the concentric phase and subsequently jumped higher—likely because of the greater RFD and power produced. The observed correlation between peak IMTP force and the COM velocity pattern is similar to previous research from Khamoui et al,[6] which illustrated a positive relationship between isometric peak force relative to body mass and vertical jump peak velocity. Similarly, Dos'Santos et al[5] also observed that stronger athletes (ie, greater IMTP peak force) jump higher. Although fewer studies report associations between peak IMTP force and COM velocity, it is interesting to briefly consider that the correlation results between peak IMTP force and the COM velocity pattern tend to agree better with findings in the literature than the correlation results for peak IMTP force and the GRF pattern. Collectively, results from the literature and the current study, thus, seem to indicate that greater maximal strength allows athletes to jump higher because they can move at faster velocities.

The GRF and COM velocity patterns were also correlated with each other. Notably, the first GRF PC was positively correlated with the first COM velocity PC and so on. These associations likely arose because the GRF data were used to calculate COM velocity data, and any variance explained by a GRF PC during specific phases of the GRF time series may have subsequently affected the COM velocity time series data at similar time points. For example, the correlation between the first GRF and first COM velocity PC indicates that greater decrease in GRF during the unweighing phase was associated with greater downward COM velocities during the same phase. Similarly, greater overall peak GRF (GRF PC2) was associated with greater velocities during the concentric phase of the CMJ (COM velocity PC2). Together, these results reinforce the potential benefit of waveform analysis since it provided a much clearer relationship between dynamic GRF and COM velocity patterns during CMJ and better explained how these patterns relate to maximal isometric strength than had been reported previously. Though beyond the scope of this article, one way to investigate the associations of GRF and COM velocity patterns would be with bivariate functional data analysis techniques.[18,19]

There are a few limitations of the current study that should be noted. First, the participants of this study consisted only of female lacrosse players. Whether the observed relationships between strength and CMJ patterns exist in other sports, or male athletes, needs to be investigated in future studies. Second, the current study used only a cross-sectional experimental design, and therefore, it remains to be determined whether the observed relationships change over the course of a competitive season or with specific training interventions. Finally, the current study applied linear normalization to the time series data. Given that different normalization methods exist,[20,21] it may be possible that application of these methods would lead to different results. That said, linear normalization of the CMJ was chosen because it enables easy interpretation for practitioners and retains the relative structure between temporal events (eg, percentage duration of specific phases). This last item also underscores the importance of proper interpretation of PC scores that capture temporal variations (eg, phase shifts) in the data and suggests that any related conclusions should be limited to the relative timing of events and not the absolute duration of a given CMJ phase.

Practical Applications

An athlete's ability to produce force during the IMTP is associated with the dynamic movement patterns they exhibit during CMJ. Stronger athletes exhibited CMJ patterns that were characterized by greater eccentric RFD and greater overall peak GRF. In addition, during the concentric phase of the CMJ, stronger athletes also exhibited a more unimodal GRF profile and greater COM velocities. Interestingly, similar findings were not observed when correlations between maximal strength and discrete CMJ variables were analyzed, which suggests a loss of important and relevant information when reducing time series data and analyzing only discrete variables. Practitioners should consider using PCA to extract and analyze dynamic patterns from CMJ data over discrete variables if they are interested in obtaining better understanding of movement dynamics in relation to maximal strength and potentially monitoring these factors over time or in response to training interventions, and so on.

The results of the current study may be used to frame specific directional hypotheses for future investigations on CMJ movement patterns as many questions arise from the findings. For example, 2 training-related examples would be:

1. A training-related increase in relative maximal strength will result in CMJ patterns that exhibit a more unimodal (vs bimodal) GRF profile.

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2. A loss in relative maximal strength due to detraining will result in a COM velocity pattern characterized by diminished magnitudes during the last 25% of the CMJ.

The use of PCA in applied settings may, therefore, provide practitioners with a valuable method for identifying athlete-specific movement patterns, especially as they relate to acute or longitudinal changes in performance due to either training, fatigue, or feedback.

Conclusions

Maximal strength is correlated with specific GRF and COM velocity patterns during the CMJ in female collegiate athletes. Stronger players showed GRF time series patterns characterized by greater eccentric phase RFD, greater peak GRF, and a unimodal GRF profile. Furthermore, stronger athletes exhibited COM velocity patterns characterized by higher COM velocities during the concentric phase of the CMJ. Importantly, although GRF and COM velocity patterns correlated with discrete parameters of CMJ performance, none of the discrete parameters was related to maximal isometric strength. These results emphasize that PCA may provide better information about the relationship between maximal strength and CMJ dynamics than discrete variables and is, thus, likely to be more beneficial for coaches and sport science researchers.

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